# The Subtractive Cosmological Model:

# Accelerated Expansion as a Function of Baryonic Dark Matter Sequestration

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**ABSTRACT**

This paper proposes a novel subtractive model of the Universe, suggesting that Dark Energy (DE) is not a fundamental repulsive force, but rather a perceived effect resulting from the localised depletion of a pre-existing, high-viscosity Dark Matter (DM) medium. We hypothesise that the Universe transitioned from a saturated state post-inflation — where DM acted as a universal gravitational brake — to an accelerated state as baryonic structures (Supermassive Black Holes and Galactic Nuclei) began to sequester this medium. By treating Dark Matter as a self-interacting superfluid with internal tension, we demonstrate that the scooping of this mass into baryonic singularities creates low-viscosity voids in the cosmic web. These regions of depleted density allow for the natural, unimpeded expansion of the vacuum, mimicking the effects of a cosmological constant ($\Lambda$). We further explore the Yin-Yang relationship between matter types, arguing that the Missing Satellites Problem and the Cusp-Core Problem are direct evidence of this ongoing depletion. This model offers a zero-sum alternative to standard $\Lambda$CDM cosmology, aligning the onset of accelerated expansion with the peak of baryonic structure formation. A novel observational prediction is introduced: a redshift-dependent gravitational lensing attenuation testable with current JWST survey data.

# 1. Introduction

The ΛCDM concordance model has proven remarkably successful in describing the large-scale structure of the Universe, the anisotropies of the Cosmic Microwave Background (CMB), and the observed abundance of light elements (Planck Collaboration, 2018). Central to this framework is the postulate that the Universe's energy budget is dominated by two poorly understood components: Cold Dark Matter (CDM), comprising approximately 27%, and Dark Energy (Λ), comprising approximately 68% of the total energy density (Planck Collaboration, 2018). Despite its predictive power, ΛCDM carries several unresolved tensions that motivate the exploration of alternative frameworks.

## *1.1 The Cosmological Constant Problem*

The most acute theoretical difficulty facing ΛCDM is the cosmological constant problem. Quantum field theory predicts a vacuum energy density that exceeds the observed value of Λ by approximately 120 orders of magnitude (Weinberg, 1989; Carroll, 2001). This represents the largest known discrepancy between theoretical prediction and observation in physics. While numerous mechanisms have been proposed to account for this mismatch — including supersymmetry, the string landscape, and anthropic selection arguments — no consensus solution has emerged (Padmanabhan, 2003). This suggests that the physical interpretation of Λ as vacuum energy may itself be flawed, and that the observed acceleration may arise from a different mechanism entirely.

## *1.2 The Coincidence Problem*

A related difficulty is the so-called coincidence problem: why does the energy density of Dark Energy only come to dominate the Universe at late cosmic times, approximately 5–6 billion years ago, at a redshift of $z \approx 0.4$ (Riess et al., 1998; Perlmutter et al., 1999)? In ΛCDM, Λ is a fixed constant, yet its dominance over matter density is a relatively recent development in cosmic history. The near-equality of dark matter and dark energy densities at the present epoch — despite their fundamentally different scaling with the expansion factor — appears highly fine-tuned and lacks a natural dynamical explanation (Zlatev et al., 1999). Any viable alternative to ΛCDM must account for this temporal coincidence without invoking fine-tuning.

## *1.3 The Hubble Tension*

A further observational tension has emerged from discrepant measurements of the present-day expansion rate $H_0$. Early-universe probes, particularly CMB observations from the Planck satellite, yield $H_0 = 67.4 \pm 0.5$ km $s^{-1}$ $Mpc^{-1}$ (Planck Collaboration, 2018), while late-universe distance ladder measurements consistently return higher values, with $H_0 = 73.0 \pm 1.0$ km $s^{-1}$ $Mpc^{-1}$ (Riess et al., 2022). This ~5σ discrepancy has persisted across independent methodologies and datasets, suggesting it may reflect a genuine inadequacy in the standard model rather than systematic error (Verde et al., 2019). A physical model in which the expansion rate is not globally uniform — but instead varies as a function of local matter density history — would naturally accommodate this tension.

## *1.4 Dark Matter as a Self-Interacting Medium*

Separately, the standard collisionless CDM paradigm faces challenges on small scales. N-body simulations predict cuspy dark matter density profiles at galactic centres and a large number of satellite subhalos, yet observations consistently reveal cored profiles and fewer satellites than predicted — the so-called Cusp-Core and Missing Satellites problems respectively (Moore et al., 1994; Klypin et al., 1999). These discrepancies have motivated proposals in which dark matter is not purely collisionless, but instead exhibits self-interaction (Spergel & Steinhardt, 2000) or superfluid behaviour (Berezhiani & Khoury, 2015). In the superfluid dark matter framework, the DM condensate supports phonon-mediated forces and collective fluid behaviour, producing dynamics that differ substantially from the pressureless dust approximation of CDM.

### *1.5 This Work*

In this paper, we propose a Subtractive Cosmological Model (SCM) in which the observed acceleration of cosmic expansion arises not from a fundamentally repulsive dark energy component, but from the progressive depletion of a pre-existing dark matter medium through baryonic sequestration. We treat the dark matter as a self-interacting superfluid — consistent with the Bose-Einstein Condensate (BEC) dark matter framework — and hypothesise that the gravitational collapse of baryonic matter into Active Galactic Nuclei (AGN) and Supermassive Black Holes (SMBHs) acts to concentrate and relocate this medium from a diffuse cosmological field into localised galactic structures.

Under this framework, Dark Energy is reinterpreted not as an independent energy component, but as the observational signature of a declining global dark matter density. The cosmological constant $\Lambda$ is reconsidered as an effective parameter quantifying the rate of this depletion rather than a property of the vacuum itself. We further propose that the temporal coincidence problem is resolved naturally, as accelerated expansion emerges only once baryonic structure formation reaches sufficient scale to meaningfully deplete the global DM field — consistent with the observed onset of acceleration at $z \approx 0.4$, coinciding with the peak epoch of SMBH growth.

The remainder of this paper is structured as follows. Section 2 describes the physical properties of the proposed dark matter superfluid and the mechanics of baryonic sequestration. Section 3 presents the modified expansion framework. Section 4 identifies existing observational evidence consistent with the model, including a novel predicted signature in redshift-dependent gravitational lensing attenuation. Section 5 discusses outstanding challenges and directions for future work.

## 2. The Dark Matter Superfluid

### *2.1 Theoretical Basis*

The possibility that dark matter behaves as a superfluid has been explored in the context of Bose-Einstein Condensate (BEC) dark matter, in which ultra-light bosonic particles occupy a coherent quantum ground state at cosmological densities (Hu et al., 2000; Berezhiani & Khoury, 2015). In this framework, dark matter is not a gas of collisionless particles but a coherent quantum fluid characterised by a macroscopic wavefunction. The fluid supports collective excitations — phonons — and exhibits the defining properties of superfluidity: zero entropy, quantised vorticity, and resistance to perturbation below a critical velocity.

This framework has been invoked primarily to explain the observed flat rotation curves of galaxies and the small-scale structure problems of CDM (Berezhiani & Khoury, 2015; Ferreira et al., 2019). We adopt the superfluid description here but extend its implications to the global cosmological medium, treating the dark matter as a continuous superfluid field permeating the full volume of the Universe rather than as a component localised within galactic halos.

### *2.2 Physical Properties of the DM Superfluid*

We characterise the dark matter superfluid by the following properties relevant to the SCM:

**Internal tension and resistance to inflow.** Unlike a classical Newtonian fluid, a superfluid does not flow to fill a cavity in the absence of an applied pressure gradient exceeding a critical threshold. This property — analogous to the yield stress of a viscoplastic material — means that voids created by baryonic sequestration are not refilled by the surrounding medium on cosmological timescales. This behaviour is consistent with the observed persistence of large-scale cosmic voids and the stability of underdense regions across cosmic time (Pan et al., 2012).

**Quantised vorticity.** When a rotating or moving mass is introduced into a superfluid, the fluid cannot develop continuous rotational flow. Instead, angular momentum is accommodated through the formation of discrete quantised vortices (Feynman, 1955). In the cosmological context, gravitationally collapsing baryonic structures induce vortex formation in the surrounding DM superfluid. The cores of these vortices represent localised regions of elevated DM density, while the surrounding medium is depleted.

**Zero bulk viscosity.** In the absence of external perturbation, the superfluid flows without dissipation. This property explains the large-scale uniformity of the dark matter distribution at early epochs, consistent with CMB observations (Planck Collaboration, 2018), while permitting localised depletion as baryonic structures develop.

**Coherence length.** The superfluid is characterised by a coherence length $\xi$ below which quantum pressure prevents further compression. This sets a minimum scale for DM clumping and is consistent with the observed suppression of small-scale structure relative to CDM predictions (Hu et al., 2000).

### *2.3 The Primordial Saturation Hypothesis*

We introduce, as a working hypothesis, the possibility that the dark matter superfluid preceded the epoch of baryonic matter formation. Under this hypothesis, the pre-inflationary or immediately post-inflationary Universe consisted of a spatially saturated DM superfluid at maximum coherent density $\rho_{DM,i}$. The introduction of baryonic matter — whether through inflationary reheating or a distinct phase transition — injected mass-energy into this pre-existing medium, acting as an impurity field within the superfluid.

We note that this hypothesis is not required for the central claims of the SCM, which concern the late-time depletion of the DM field and its cosmological consequences. However, it provides a natural initial condition for the model and offers an alternative interpretation of the primordial density fluctuations observed in the CMB as the imprint of baryonic impurities displacing the superfluid during the inflationary epoch, rather than purely quantum fluctuations in a vacuum field. We treat this element of the model as speculative and defer its formal development to future work.

## 3. Baryonic Sequestration

### *3.1 Definition and Mechanism*

We define *baryonic sequestration* as the process by which gravitational collapse of baryonic matter removes dark matter from the global cosmological superfluid field and concentrates it into localised, virialized structures. This process operates through two primary channels:

**Direct accretion.** As baryonic mass concentrates into a gravitational potential well, the surrounding DM superfluid falls within the well's capture radius. A fraction of this DM crosses the Schwarzschild radius of the central black hole and is incorporated into the singularity. The gravitational mass of the absorbed DM is conserved within the black hole; however, it is removed from the distributed superfluid field and its gravitational influence is now exerted exclusively from the black hole's position.

The combined effect of both channels is a progressive reduction in the density of the global DM superfluid field — the component of the DM distribution that provides large-scale gravitational coherence across the cosmic web.

**Orbital localisation.** DM not directly accreted but captured within the gravitational potential of the galaxy forms a bound halo. This DM is no longer part of the global cosmological superfluid — it is virialized within the galaxy's gravitational well and does not contribute to the large-scale field tension of the inter-halo medium. We refer to this as *passive sequestration*, distinct from active accretion into black holes.

### *3.2 Gravitational Topology and the Field Deficit*

A critical distinction in the SCM is between *total gravitational mass* and *gravitational field topology*. Mass conservation ensures that the total gravitational mass of the Universe is unchanged by sequestration. However, the spatial distribution of that mass is fundamentally altered: a diffuse superfluid field exerts gravitational influence uniformly across its volume, while the same mass concentrated into point sources exerts gravity that falls off as $r^{-2}$ from each source.

Consider a region of inter-halo space previously permeated by a uniform DM superfluid at density $\rho_{DM}$. The gravitational potential at a point within this region receives contributions from all surrounding mass elements. As sequestration proceeds, those mass elements are relocated to galactic halos and black holes at distance d from the region. The contribution of the sequestered mass to the local potential now scales as $d^{-1}$ rather than as a distributed field integral, resulting in a net reduction of the gravitational potential in inter-halo regions even though no mass has been destroyed.

We term this the *global density deficit* $\Delta\rho$, defined as:

$$\Delta\rho(t) = \rho_{DM,i} - \rho_{DM,field}(t)$$

where $\rho_{DM,i}$ is the initial field density and $\rho_{DM,field}(t)$ is the density of the unsequestered, distributed superfluid at time t. It is $\Delta\rho$, not the total DM mass, that governs the large-scale gravitational coherence of the cosmic medium.

### *3.3 The Sequestration Rate*

The rate at which the global density deficit grows is governed by the sequestration rate $\Gamma_{seq}(t)$, which we expect to track the cosmic star formation rate and AGN activity history. Observationally, both the cosmic star formation rate density and the SMBH accretion rate peak at redshifts $z \approx 1–3$, the epoch known as cosmic noon (Madau & Dickinson, 2014; Ueda et al., 2014).

We express the time evolution of the field density as:

$$\rho_{DM,field}(t) = \rho_{DM,i} - \int_0^t \Gamma_{seq}(t')\, dt'$$

where $\Gamma_{seq}(t')$ is proportional to the cosmic star formation rate density and AGN luminosity function at each epoch. The integral represents cumulative sequestration. This formulation predicts that $\rho_{DM,field}(t)$ decreases monotonically with time, with the rate of decrease accelerating during cosmic noon and declining as AGN activity subsides at lower redshifts.

Crucially, the effects of this depletion on cosmic expansion become observable only once the cumulative deficit $\Delta\rho$ is sufficient to meaningfully reduce the large-scale gravitational coherence of the medium. This introduces a natural time delay between the peak of sequestration activity and the onset of observable accelerated expansion — consistent with the observed acceleration beginning at $z \approx 0.4$, approximately 3–4 Gyr after the peak of cosmic noon.

### *3.4 Black Hole Motion and Asymmetric Depletion*

The spatial pattern of DM depletion is further governed by the motions of black holes and galactic structures through the superfluid medium. Black holes do not move preferentially toward or away from voids; their trajectories are determined by gravitational attraction to nearby mass concentrations. However, because the DM superfluid does not flow to refill depleted regions — owing to its resistance to inflow below the critical pressure threshold — the regions through which black holes have passed accumulate a persistent density deficit.

The velocity of a moving black hole relative to the local DM superfluid determines the geometry of sequestration. A black hole moving supersonically relative to the local sound speed of the superfluid creates a Bondi-Hoyle-Lyttleton accretion geometry (Edgar, 2004), with a characteristic gravitational focusing radius:

$$r_{BHL} = 2GM_{BH} / (v^2 + c_s^2)$$

where v is the relative velocity between the black hole and the superfluid, and $c_s$ is the superfluid sound speed. DM within this radius is captured; DM outside it is deflected but not accreted. This creates an elongated depletion wake behind the moving black hole — a persistent low-density trail in the superfluid. The superfluid's resistance to inflow ensures this wake is maintained rather than refilled.

## 4. Modified Expansion Framework

### *4.1 Modified Friedmann Equations*

In standard ΛCDM, the Friedmann equation governing the expansion rate H(t) is:

$$H^2(t) = (8\pi G/3)(\rho_m + \rho_r + \rho_\Lambda)$$

where $\rho_m$, $\rho_r$, and $\rho_\Lambda$ are the energy densities of matter, radiation, and the cosmological constant respectively, and $\rho_\Lambda = \Lambda/8\pi G$ is treated as a fixed property of the vacuum.

In the SCM, we replace the static cosmological constant with a dynamic term arising from the global density deficit:

$$H^2(t) = (8\pi G/3)(\rho_b + \rho_{DM,field}(t) + \rho_r) + (8\pi G/3)f(\Delta\rho(t))$$

where $\rho_b$ is the baryonic matter density, $\rho_{DM,field}(t)$ is the evolving field DM density as defined in Section 3.2, and $f(\Delta\rho(t))$ is a function encoding the contribution of the density deficit to the effective expansion rate.

We propose that $f(\Delta\rho)$ represents the release of gravitational tension in the superfluid as the field density drops. In the superfluid framework, the coherent medium exerts a restoring force — analogous to surface tension — that resists expansion. As $\Delta\rho$ grows, this restoring force diminishes. The effective expansion rate is therefore the result not of a new repulsive energy being added, but of the gradual removal of a restraining term. Formally:

$$f(\Delta\rho(t)) \approx -(\rho_{DM,i} - \rho_{DM,field}(t)) \cdot \alpha$$

where α is a dimensionless coupling constant relating the density deficit to the reduction in gravitational tension. In the limit where $\rho_{DM,field} \to 0$, the restraining term vanishes entirely and the Universe expands at a rate governed solely by its baryonic and radiation content, plus the residual momentum of the initial expansion event.

The determination of α from first principles requires a fully relativistic treatment of superfluid field theory in an expanding spacetime, which we defer to future work. Here we note that α is constrained observationally: it must reproduce the observed onset of acceleration at $z \approx 0.4$ and the present-day expansion rate $H_0$, consistent with Type Ia supernova observations (Riess et al., 1998; Perlmutter et al., 1999).

### *4.2 The Effective Cosmological Constant*

In the SCM, the cosmological constant $\Lambda_{eff}$ is not a fixed vacuum property but a time-evolving quantity:

$$\Lambda_{eff}(t) = 8\pi G \cdot \alpha \cdot \Delta\rho(t)$$

At early times, $\Delta\rho \approx 0$ (negligible sequestration has occurred) and $\Lambda_{eff} \approx 0$, consistent with the matter-dominated expansion of the early Universe. At late times, as cumulative sequestration grows, $\Lambda_{eff}$ increases, producing the observed acceleration. In the far future, as $\rho_{DM,field} \to 0$, $\Lambda_{eff}$ asymptotes to a maximum value determined by the initial field density $\rho_{DM,i}$.

This formulation resolves the cosmological constant problem by construction: $\Lambda_{eff}$ is not derived from vacuum energy and therefore carries no quantum field theory prediction. Its value is set by the history of baryonic structure formation, which is a classical gravitational process.

### *4.3 The Coincidence Problem*

The SCM provides a natural, dynamical resolution to the coincidence problem. In ΛCDM, the near-equality of the matter and dark energy densities at the present epoch is a numerical coincidence with no causal explanation. In the SCM, the dominance of the effective dark energy term over the matter density at late times is causally linked to the history of baryonic structure formation.

Specifically, accelerated expansion becomes observationally significant only once the cumulative sequestration integral $\int \Gamma_{seq}\, dt'$ is large enough to drive a meaningful density deficit. This condition is satisfied only after the peak of cosmic noon — after sufficient SMBHs and galactic structures have formed to deplete the global field. The ~3–4 Gyr delay between the peak sequestration rate ($z \approx 2$) and the onset of acceleration ($z \approx 0.4$) is therefore a direct consequence of the time required for cumulative sequestration to reach a critical threshold.

In this picture, it is not a coincidence that acceleration began when it did. It began precisely when baryonic activity had been sustained long enough and at sufficient scale to meaningfully erode the gravitational coherence of the dark matter medium.

### *4.4 The Hubble Tension*

The SCM predicts that the expansion rate H is not globally uniform but varies as a function of the local dark matter field density $\rho_{DM,field}$. Regions of the Universe where sequestration has been more efficient — due to higher concentrations of SMBHs and star-forming galaxies — will have a lower local $\rho_{DM,field}$ and therefore a higher local effective expansion rate.

This introduces a systematic inhomogeneity in H that depends on the integrated baryonic activity history of each region. Early-universe probes such as the CMB sample the expansion history averaged over the full observable volume, weighted toward the primordial epoch when sequestration was negligible and $\rho_{DM,field}$ was near its initial value. Late-universe probes such as Type Ia supernovae and Cepheid distance ladders sample the local expansion rate in our cosmological neighbourhood, where several billion years of baryonic activity have produced a measurable density deficit.

The SCM therefore predicts that early-universe measurements of $H_0$ should systematically underestimate the present-day local expansion rate relative to late-universe measurements — consistent with the observed Hubble tension. The magnitude of the discrepancy is set by the local density deficit, which is a function of the baryonic activity history of our local cosmological volume.

## 5. Observational Evidence and Predictions

### *5.1 The Cusp-Core Problem*

N-body simulations of collisionless CDM consistently produce dark matter halos with cuspy central density profiles scaling as $\rho \propto r^{-1}$ (NFW profile; Navarro, Frenk & White, 1997), while observations of dwarf galaxies routinely reveal flat, cored profiles (Oh et al., 2011; Read et al., 2019). In ΛCDM, this discrepancy is attributed to baryonic feedback — supernova-driven outflows that redistribute dark matter outward (Pontzen & Governato, 2012).

In the SCM, the cusp-core problem receives an alternative interpretation. The formation of baryonic structures within a DM superfluid halo initiates sequestration: the gravitational collapse of stars and the activation of AGN create quantised vortices and accretion-driven depletion in the halo's central region. The result is a reduction in central DM density that mimics the cored profiles observed, but through sequestration rather than outflow. This mechanism operates even in systems where supernova feedback is insufficient — such as ultra-diffuse galaxies with low star formation rates — potentially explaining residual cusp-core discrepancies that baryonic feedback alone cannot account for.

### *5.2 The Missing Satellites Problem*

ΛCDM simulations predict an order-of-magnitude more satellite subhalos around Milky Way-mass galaxies than are observed (Moore et al., 1999; Klypin et al., 1999). In the SCM, this deficit is interpreted as evidence of early sequestration. The smallest DM subhalos — those near or below the coherence length ξ of the superfluid — are the most susceptible to disruption by baryonic activity. Early star formation and AGN activity in the proto-galactic environment would have selectively depleted these low-mass substructures, reducing the satellite count relative to simulations that do not include sequestration physics.

Furthermore, the superfluid's internal tension — its resistance to fragmentation below the coherence scale — naturally suppresses the formation of very small subhalos, providing a structural mechanism for the observed deficit independent of baryonic disruption.

### *5.3 Cosmic Voids*

Large-scale cosmic voids — regions tens to hundreds of megaparsecs across containing almost no galaxies or luminous matter — are among the most distinctive features of the cosmic web. In ΛCDM, voids are primordial underdensities that have expanded over time due to the net outward pressure of the surrounding overdense filaments and walls (Peebles, 2001).

In the SCM, voids receive a complementary interpretation: they are regions where the DM superfluid has been most depleted by the surrounding baryonic structures. The filaments and galaxy clusters bounding the voids are sites of concentrated baryonic activity — the highest densities of SMBHs and star-forming galaxies — and therefore the highest rates of sequestration. The sequestration in these bounding structures depletes the DM field not only locally but across the adjacent void volume, as the superfluid field tension is reduced throughout the region.

This interpretation predicts that voids should exhibit the fastest local expansion rates — consistent with observations showing that voids expand faster than the cosmic mean (Hamaus et al., 2014) — because they are the regions of lowest $\rho_{DM,field}$ and therefore highest $\Lambda_{eff}$. Importantly, the SCM predicts this as a causal consequence of sequestration rather than an initial condition, which may be distinguishable from the ΛCDM prediction through the void expansion rate history at different redshifts.

### *5.4 Redshift-Dependent Gravitational Lensing Attenuation*

We introduce a novel observational prediction of the SCM. Gravitational lensing measurements integrate the total mass along the line of sight between observer and source. This includes contributions from the DM superfluid field as well as sequestered (halo and black hole) DM. In ΛCDM, the total DM mass is conserved and lensing strength for a given mass configuration does not evolve systematically with cosmic time beyond the effects of the expanding geometry.

In the SCM, the field component of the DM density — which contributes to line-of-sight lensing — decreases monotonically with time as sequestration proceeds. We therefore predict a *redshift-dependent lensing attenuation*: the lensing convergence κ for a given projected baryonic mass should be systematically lower at low redshift than at high redshift, after accounting for geometric factors. Formally:

$$\kappa(z) \propto \Sigma_{DM,field}(z) + \Sigma_{DM,halo}(z) + \Sigma_{b}(z)$$

where Σ denotes projected surface density. In ΛCDM, the total $\Sigma_{\mathrm{DM}}$ is approximately conserved; in the SCM, $\Sigma_{\mathrm{DM,field}}$ decreases with decreasing redshift while $\Sigma_{\mathrm{DM,halo}}$ increases. The net effect on κ depends on the ratio of field to halo DM along the line of sight. For lines of sight passing through underdense regions — cosmic voids or inter-filament space — the halo contribution is minimal and the field depletion dominates, predicting a measurable weakening of lensing convergence at low redshift.

**Testable prediction:** For galaxy clusters of equivalent baryonic mass at different redshifts, the SCM predicts weaker lensing convergence at lower redshift, with the magnitude of attenuation correlating with the integrated AGN and star formation activity history of the surrounding cosmological volume. This prediction is distinguishable from ΛCDM, which predicts no such systematic attenuation beyond geometric corrections.

**Proposed dataset:** The JWST PEARLS, COSMOS-Web, and CEERS programs provide deep imaging across multiple redshift bins with sufficient lensing statistics to test this prediction. A dedicated analysis comparing weak lensing convergence profiles of mass-matched clusters across redshift, controlling for baryonic mass and local environment, would constitute a direct test of the SCM's lensing attenuation prediction.

### *5.5 Dark Matter Deficient Galaxies*

The discovery of galaxies with anomalously low dark matter content — such as NGC 1052-DF2 and NGC 1052-DF4 (van Dokkum et al., 2018, 2019) — poses a challenge to standard ΛCDM, in which all galaxies should reside within substantial DM halos. In the SCM, dark matter deficient galaxies represent an extreme endpoint of sequestration: systems in which the local DM field has been so thoroughly depleted — whether by the galaxy's own baryonic activity or by the proximity of a more massive neighbouring system — that the residual halo DM is insufficient to dominate the galaxy's dynamics.

The observed association of NGC 1052-DF2 and DF4 with the massive elliptical NGC 1052 is consistent with this interpretation: the dominant neighbour's sequestration activity may have depleted the shared DM field in the region, starving the satellite galaxies of their halos.

## 6. Discussion

### *6.1 Relationship to Existing Alternative Models*

The SCM shares conceptual ground with several existing frameworks while differing in key respects.

**Superfluid dark matter (Berezhiani & Khoury, 2015)** treats DM as a BEC superfluid within galactic halos to reproduce MOND-like dynamics. The SCM extends this framework to the global cosmological field and introduces the sequestration mechanism as a driver of late-time acceleration, which is absent from the Berezhiani-Khoury model.

**Interacting dark energy models (Wetterich, 1995; Amendola, 2000)** allow energy exchange between dark matter and dark energy. The SCM differs in that it does not require a separate dark energy field — the effective dark energy arises purely from the depletion of the dark matter field, requiring no additional degrees of freedom.

**Running vacuum models (Lima & Trodden, 1996; Solà et al., 2019)** treat Λ as a time-varying quantity. The SCM provides a specific physical mechanism — baryonic sequestration — that drives the time variation of the effective cosmological constant, grounding it in observable astrophysical processes rather than phenomenological parameterisation.

**Backreaction models (Buchert, 2000)** propose that the observed acceleration arises from the backreaction of inhomogeneities on the average expansion rate. The SCM is compatible with the general spirit of this approach but differs in attributing the inhomogeneity to DM field depletion rather than purely geometric effects.

### *6.2 Outstanding Challenges*

We identify the following as the primary theoretical and observational challenges facing the SCM:

**Quantitative framework.** The coupling constant α relating the density deficit to the effective expansion rate has not been derived from first principles in this work. A full treatment requires a relativistic quantum field theory of the DM superfluid in an expanding spacetime — specifically, a BEC field theory on a Friedmann-Lemaître-Robertson-Walker (FLRW) background. This is mathematically non-trivial but falls within the existing literature on quantum fields in curved spacetime (Parker & Toms, 2009).

**CMB power spectrum compatibility.** The acoustic oscillation peaks in the CMB are exquisitely well reproduced by ΛCDM with a collisionless CDM component. A superfluid DM medium with internal pressure and self-interaction would modify the sound horizon and damping scale of the baryon-photon fluid in ways that must be quantified. In particular, the superfluid sound speed $c_s$ must be sufficiently small at the epoch of recombination that it does not dampen the observed oscillation pattern. This is not ruled out a priori — ultra-light BEC dark matter models have been shown to be compatible with CMB observations for sufficiently large boson masses (Hu et al., 2000) — but requires explicit calculation for the SCM's parameter space.

**Big Bang Nucleosynthesis (BBN).** The standard BBN framework constrains the total energy density of relativistic species at $z \sim 10^8$. If the DM superfluid contributes a pressure term during this epoch, it could alter the expansion rate and modify primordial element abundances. This places constraints on the equation of state of the superfluid at early times that must be satisfied.

**Void DM density.** The SCM predicts that cosmic voids should be depleted in field DM. Independent measurements of DM density in voids — through weak lensing of void interiors — show that voids are underdense in DM relative to the cosmic mean (Clampitt & Jain, 2015). While this is qualitatively consistent with the SCM, it is also predicted by ΛCDM as a consequence of the primordial density field. Distinguishing between these two explanations requires measurements of the *evolution* of void DM density over redshift, which is a prediction unique to the SCM.

**Gravitational wave constraints.** The detection of gravitational waves from binary black hole and neutron star mergers by LIGO/Virgo provides a new probe of the DM density in the local environment of compact objects. If the DM superfluid is present at the densities required by the SCM, it may produce detectable phase shifts or damping in gravitational wave signals propagating through the medium (Morisaki & Soda, 2018). Current constraints may already limit the local DM superfluid density, and future observations with LISA will provide much stronger bounds.

### *6.3 Directions for Future Work*

We identify the following as priorities for the formal development of the SCM:

1. **Derivation of α from BEC field theory** on an FLRW background, producing a first-principles relationship between the sequestration rate and the effective expansion rate.

2. **N-body simulations incorporating sequestration** — modifying existing SIDM simulation codes (Robertson et al., 2019) to include a time-evolving field density component that depletes in proportion to local baryonic activity.

3. **Lensing attenuation analysis** using JWST survey data, testing the redshift-dependent convergence prediction of Section 5.4 against mass-matched cluster samples.

4. **Sequestration rate function calibration** using the observed cosmic star formation rate density and AGN luminosity function to constrain $\Gamma_{seq}(t)$ and test whether the resulting $\Lambda_{eff}(t)$ reproduces the observed expansion history.

5. **CMB compatibility analysis** to determine the range of superfluid boson masses and self-interaction cross sections consistent with the observed acoustic oscillation spectrum.

## 7. Conclusion

We have presented the Subtractive Cosmological Model, a framework in which the observed acceleration of cosmic expansion is interpreted not as the consequence of a fundamental repulsive dark energy component, but as the progressive removal of a gravitational restraining force — the dark matter superfluid field — through baryonic sequestration.

The central claims of the model are as follows:

1. Dark matter behaves as a self-interacting superfluid on cosmological scales, characterised by internal tension and resistance to cavity infill.

2. The formation of baryonic structures — SMBHs, AGN, and galactic halos — progressively sequesters dark matter from the global cosmological field into localised, virialized configurations, reducing the field density monotonically over cosmic time.

3. This reduction in field density decreases the large-scale gravitational coherence of the cosmic medium. The observed accelerated expansion is the signature of this declining coherence — not of a new energy component.

4. The effective cosmological constant $\Lambda_{eff}$ is a time-evolving function of the cumulative density deficit, resolving the cosmological constant problem by eliminating its vacuum energy interpretation.

5. The onset of acceleration at $z \approx 0.4$ is a natural consequence of the time required for cumulative sequestration to reach a critical threshold following the peak of cosmic noon, resolving the coincidence problem without fine-tuning.

6. The Hubble tension arises naturally from the inhomogeneous distribution of sequestration: regions of higher baryonic activity have lower local field densities and higher effective expansion rates, producing a systematic offset between early- and late-universe measurements of $H_0$.

7. The model predicts a novel observational signature — redshift-dependent gravitational lensing attenuation — that is distinguishable from ΛCDM predictions and testable with existing JWST survey data.

The SCM does not require new fundamental forces, additional energy fields, or fine-tuned initial conditions. It proposes that the apparent darkness of both dark matter and dark energy reflects two aspects of a single evolving medium: what we call dark energy is the Universe's gradually failing grip on itself, as the structural integrity of the cosmic medium is consumed by the very structures it allowed to form.

Formal quantitative development — particularly the derivation of the coupling constant α and the demonstration of CMB compatibility — remains outstanding and is the subject of ongoing work.

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